Poker Exercises 1

- A poker hand contains 5 out of 52 cards. The cards are 13 kinds (1-Ace, 2,..,10,J,Q,K) and 4 suites (spades, hearts, clubs, diamonds)
- What is the total number of poker hands?
- C(52,5)
- How many hands contain the Ace of Spades and the King of Spades?
- Since these 2 cards are fixed, I have C(50,3) choices for the remaining 3 cards.
- How many hands contain exactly 1 Ace?
- There are 4 ways to choose the Ace. Once the Ace is selected, there are C(48,4) ways to choose the remaining four cards: $4 \cdot C(48,4)$

Poker Exercises 2

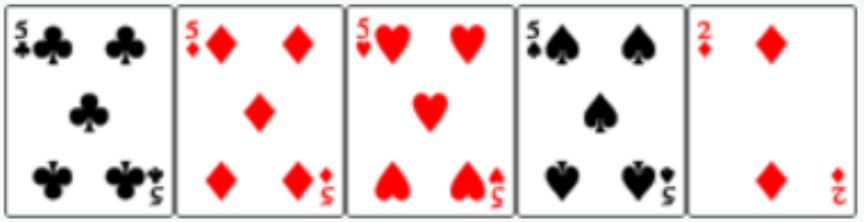
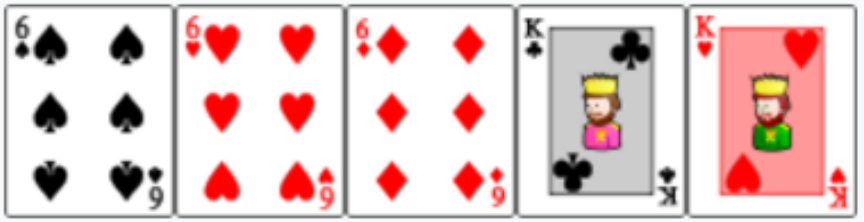
- A poker hand contains 5 out of 52 cards. The cards are 13 kinds (1-Ace, 2,..,10,J,Q,K) and 4 suites (spades, hearts, clubs, diamonds)
- How many hands are royal flush? Royal flush is {10,J,Q,K,Ace} of the same suit. This is the strongest card in poker.
- There are 4 royal flushes/one per suit
- How many hands are straight flush? A straight flush is a hand with all five cards of the same suit in sequential order starting at 1,2,….,9. 10 is excluded because it would lead to a Royal flush.
- I have C(9,1)=9 options for the beginning of straight flush, and 4 options for the suit, yielding a total of $4 \cdot 9 = 36$. There is a single option for each of the remaining cards.
- How many hands are straight? A straight is a hand with all five cards in sequential order. At least two cards are of different suit; otherwise, it would be a royal or straight flush.
- I will first count all straights including straight and royal flushes.
- I have C(10,1)=10 options for the beginning of the straight. For each of the five cards, there are 4 suite options. Therefore, total number of straights is: $10 \cdot 4^5$.
- If I exclude the straight and royal flushes, the number becomes: $10 \cdot 4^5 - 40$.

Poker Exercises 3

- A poker hand contains 5 out of 52 cards. The cards are 13 kinds (1-Ace, 2,..,10,J,Q,K) and 4 suites (spades, hearts, clubs, diamonds)
- How many hands are flush, i.e., a flush consists of 5 card of the same suit, excluding the cases of royal and straight flush.
- The flush is a combination of 5 out of 13 cards of the same suit. Thus, there are 4·C(13,5) flushes. After excluding the straight and the royal flashed the number is 4·C(13,5)-40
- How many hands contain 3-of a kind (e.g., 3 aces - the 4th and 5th card must be different because otherwise we would have stronger hands, to be discussed later):
- Number of ways to select the kind of the 3-of a kind C(13,1).
- Number of ways to select the suites of the 3-of a kind C(4,3).
- Number of ways to select the kind of the 4th and 5th card: C(12,2).
- Number of ways to select the suit of the 4th and 5th card: C(4,1)·C(4,1)
- Total number: C(13,1)·C(4,3)·C(12,2)· C(4,1)·C(4,1)
- How many hands contain 2 pairs (e.g., 2 aces and 2 kings - the 5th card be different):
- Number of ways to select the 2 pairs is C(13,2).
- Number of ways to select the suites of the 2 pairs C(4,2)·C(4,2).
- Number of ways to select the fifth card: 52-8=44.
- Total number: C(13,2)·C(4,2)·C(4,2)·44

Poker

Exercises 4

Four of a kind	
Full house	

- How many hands contain 4-of a kind?
- Number of ways to select the kind of the 4-of a kind C(13,1).
- Number of ways to select the 5th card: 48 (12 for the rank times 4 for the suit).
- Total number is: 13·48
- How many hands contain a full house?
- Number of ways to select the 2 kinds C(13,2).
- Number of ways to select the kind for the 3-of a kind C(2,1).
- Number of ways to select the suits for the 3-of a kind C(4,3)
- Number of ways to select the suits for the pair C(4,2).
- Total number is: C(13,2)·C(2,1)·C(4,3) ·C(2,2)
- Poker Hands: Royal Flush > Straight Flush > 4-of a kind > Full house > Flush > Straight > 3-of a kind > 2 pairs > 1 pair

Permutations with Indistinguishable Objects

- **Theorem**

The number of different permutations of n objects, where there are n_1 indistinguishable objects of type 1, n_2 indistinguishable objects of type 2, …, and n_k indistinguishable objects of type k, ($n=n_1+n_2+..+n_k$) is

$$\frac{n!}{n_1!\,n_2!\,\ldots n_k!}$$

A Corollary of the Theorem

The theorem assumes that $n_1+n_2+..+n_k=n$

What is the number of permutations with indistinguishable objects when $n_1+n_2+..+n_k=m<n$

$$\frac{n!}{n_1!\,(n-n_1)!}\cdot\frac{(n-n_1)!}{n_2!\,(n-n_1-n_2)!}\cdot\cdot\frac{(n-n_1\ldots-n_{k-2})!}{n_{k-1}!\,(n-n_1\ldots-n_{k-1})!}\cdot\frac{(n-n_1\ldots-n_{k-1})!}{n_k!\,(n-n_1\ldots-n_k)!}$$
$$=$$
$$\frac{n!}{n_1!\,n_2!\ldots n_{k-1}!\,n_k!\,(n-m)!}$$

Distributing Objects into Boxes

Many counting problems can be solved by enumerating the different ways objects can be placed into boxes where the order these objects are placed into the boxes does not matter.

The objects and the boxes can be either *Distinguishable* (labeled) or *Indistinguishable* (unlabeled).

Examples:

- D-D: How many ways to place n labeled objects in m labeled boxes
- D-I: How many ways to place n labeled objects in m unlabeled boxes
- I-D: How many ways to place n unlabeled objects in m labeled boxes
- I-I: How many ways to place n unlabeled objects in m unlabeled boxes

Distributing Objects into Boxes

- **Theorem**
  The number of ways to distribute n distinguishable objects into k distinguishable boxes so that n_i objects are placed into box i, where $n_1+n_2+..+n_k=m<n$ is equal to:

$$\frac{n!}{n_1!\,n_2!\ldots n_{k-1}!\,n_k!\,(n-m)!}$$

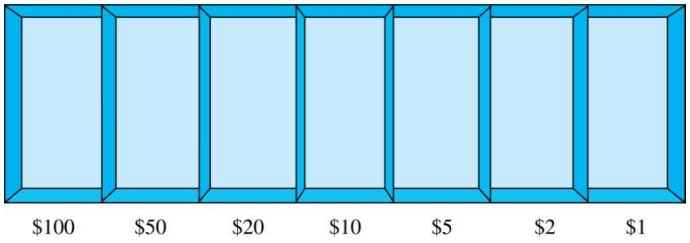
- **Special Case**
- When $n_1+n_2+..+n_k = n$ the above formula becomes:

$$\frac{n!}{n_1!\,n_2!\ldots n_{k-1}!\,n_k!}$$

- Same formulae as "Permutations with Indistinguishable Object"

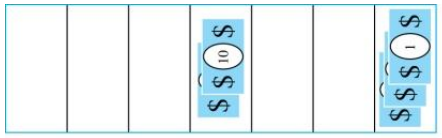
Combinations with Repetition

- How many ways are there to select 5 bills from the following 7 denominations: $1, $2, $5, $10, $20, $50, and $100? Assume there are enough bills of each denomination
- **Solution**
  Select the bills from the appropriate position of a cash box illustrated below.
- Equivalently, according to the "*Distributing Objects into Boxes*" framework, distribute 5 indistinguishable objects (bills) into 7 distinguishable boxes (denominations).

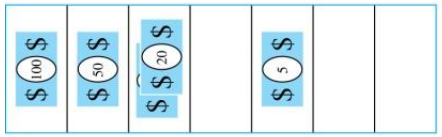


Solution (cnt'd)

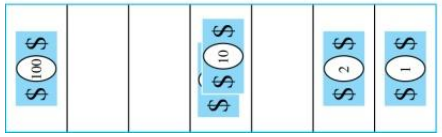
- Some possible ways of placing the five bills:



| | | | ∗ ∗ | | | ∗ ∗ ∗



∗ | ∗ | ∗ ∗ | | ∗ | |



∗ | | | ∗ ∗ | | ∗ | ∗

- # of ways to select 5 bills from 7 denominations = # permutations (with indistinguishable objects) for 6 **Bars** and 5 **Stars**.
- How many different strings of (length 11) can be made by reordering six **B**s and five **S**s?
- $C(11,5) = \frac{11!}{5!6!}$

Comments on the Bill Example

- How many ways are there to select 5 bills from 7 denominations, assuming there are enough bills of each denomination
- **An alternative solution**
  Given that there are 7 choices for the first bill, 7 choices for the second bill,…., 7 choices for the fifth bill, somebody could argue that
- The total number of ways is 7^5.
- What is wrong with this solution?
- Lets take a choice by the alternative solution : e.g., (7,7,1,1,1) means that for the first 3 bills, I select the first denomination, while for the next 2, I select the 7th.
- Any other choice, e.g., (1,7,1,1,7) with the same number of 1s and 7 is considered distinct by the alternative solution.
- But it should not be distinct because the order in which the bills are selected is not important. Bills of the same denomination are indistinguishable.
- The bars and stars representation eliminates this problem.
- All choices with three 1s and two 7s have the same representation and are counted once.
- Representation of (1,7,1,1,7) (7,7,1,1,1) and all equivalent choices using bars and stars:**||||||***

Combinations with Repetition *(bars and stars)*

- **Theorem**
  The number of r-combinations from a set with n elements when repetition of elements is allowed is $C(n+r-1, r) = \frac{(n+r-1)!}{r!(n-1)!}$
- **Proof**: Each such r-combination can be represented by a list of $n-1$ bars and r stars. The bars mark the n cells, each corresponding to a distinct element. The stars between two bars correspond to the repetitions of an element.
  The list has $n+r-1$ positions, and we choose r of them to place the stars. Or equivalently, we choose $n-1$ positions to place the bars.

Summary

TABLE 1 Combinations and Permutations With and Without Repetition.

Type	*Repetition Allowed?*	*Formula*
r-permutations	No	$\frac{n!}{(n-r)!}$
r-combinations	No	$\frac{n!}{r!\,(n-r)!}$
r-permutations	Yes	n^r
r-combinations	Yes	$\frac{(n+r-1)!}{r!\,(n-1)!}$

Example of Combinations with Repetition

- Suppose that a cookie shop has 4 different kinds of cookies. In how many different ways can 6 cookies be chosen? There are no constraints on how many cookies of each kind are chosen (e.g., all 6 may be of the same kind)
- **Solution**
  Equivalently, I wish to distribute 6 indistinguishable cookies (stars) to 4 distinguishable cookie boxes (bars).
- By previous theorem, the number of such combinations is
  $$C(4+6-1,6) = C(9,6) = C(9,3) = 84$$
- Is the solution still correct if some cookie box contains fewer than 6 cookies? (Equivalently, the cookie shop does not have 6 cookies for some cookie kind(s)).
- No. We cannot place 6 stars in that box. Solution later …
- Equivalent problem: Find the number of (nonnegative integer) solutions for equation: $x_1 + x_2 + x_2 + x_4 = 6$, where $0 \leq x_i \leq 6$ indicates the chosen number of cookies of type *i*.
- The number of solutions is $C(4+6-1,6) = C(9,6) = C(9,3) = 84$

Exercise on Equation Solutions

- How many solutions does the equation
$$x_1 + x_2 + x_3 = 5$$
have, where x_1, x_2, and x_3 are nonnegative integers?
- Equivalent problem: Suppose that a cookie shop has 3 different kinds of cookies. In how many different ways can 5 cookies be chosen?
- **Solution**
- Each solution corresponds to a way to select 5 items with repetition from 3 element types; x_1 items of type 1, x_2 of type 2, and x_3 of type 3.
- Equivalently: I wish to select 5 indistinguishable objects from 3 distinguishable boxes x_1, x_2, and x_3.
- By the previous theorem, it follows that there are
$$C(3 + 5 - 1,5) = C(7,5) = \frac{7!}{5!\,2!} = 21$$
solutions.

Exercise on Equation Solutions with minimum constraint

- How many solutions does the equation

$$x_1 + x_2 + x_3 = 5$$

have, where x_1, x_2, and x_3 are nonnegative integers and $x_3 \geq 3$?

- **Solution**
- I rewrite the equation as follows:
- $x_1 + x_2 + (x_3 - 3) = 5 - 3 \Rightarrow x_1 + x_2 + x = 2.$
- The number of solutions for the new equation is
- $C(3 + 2 - 1,2) = C(4,2) = \frac{4!}{2!2!} = 6$
- This is the number of solutions when $x \geq 0 \Rightarrow x_3 - 3 \geq 0 \Rightarrow x_3 \geq 3.$